

Teaching Exam

**Subject –
Computer
Science**

**Computational Thinking and
Programming**



Lecture No –1

By–Satyendra Chaudhary Sir

Topics to be Covered

Topic

**Computational Thinking and
Programming**



Operating System → Not all chapters covered.
→ 2h - 3 sessions

Computational Thinking & Programming ✓✓

Python

Problem $\rightarrow$ Description.

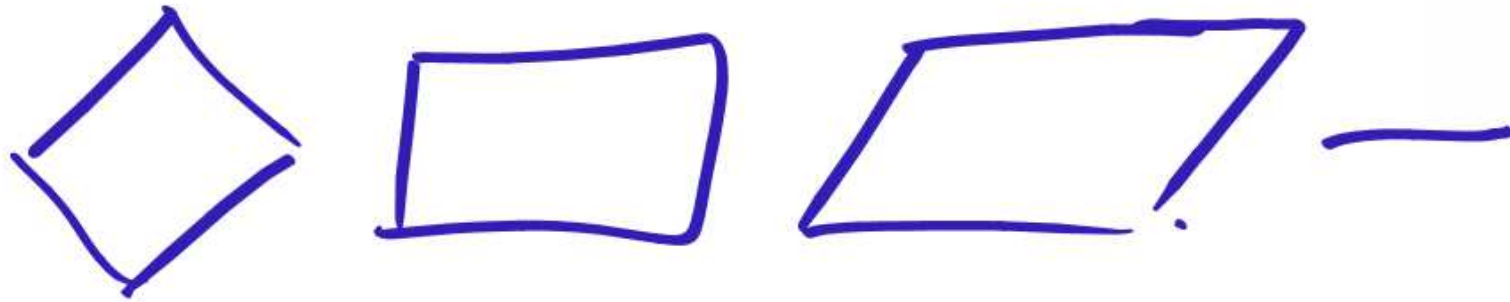
$\rightarrow$ Analysis.

Algorithm development $\rightarrow$ set of steps (rules) designed
for solving a specific problem.
Complexity

Implementation

Testing

Algorithm → Flowchart.
→ Pseudocode.

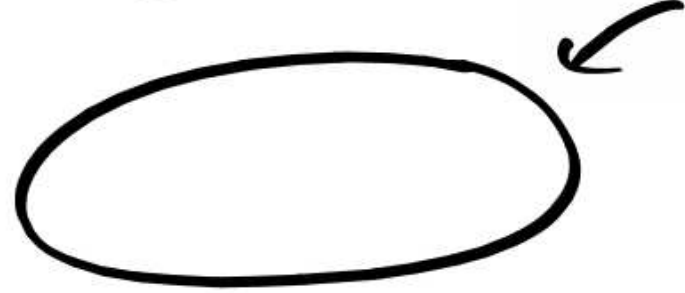


Input a, b

compute = $c = a + b$

Print = c

Flow chart



Start / End



Input / Output

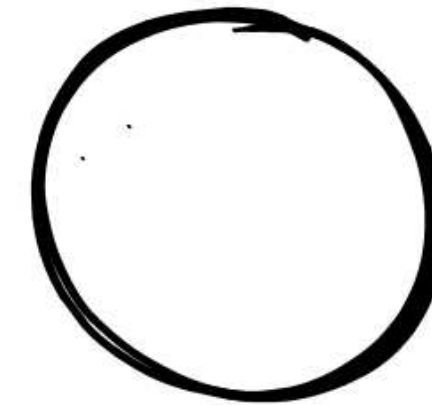
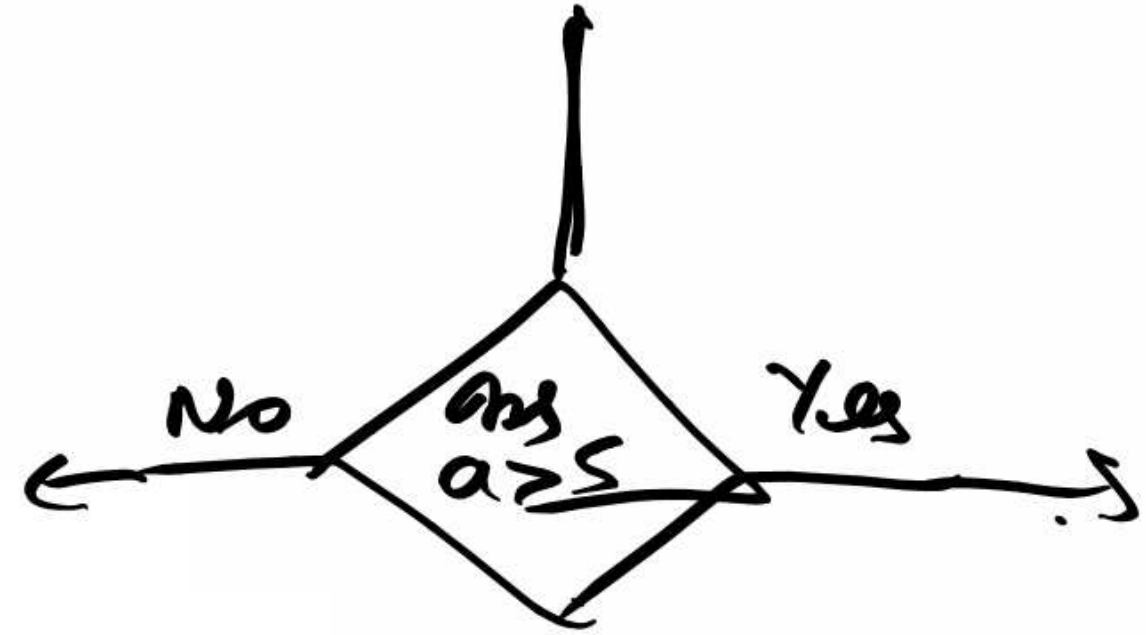


Computation
Process

→ arrow



Decision



Connector

Compute multiply with the help of addition.

$$a \times b = \underline{\hspace{2cm}}$$

$$\underline{m = a.}$$

$$a = 3$$

$$b = 5$$

$$m = 6$$

$$c = 1$$

$$m = 9$$

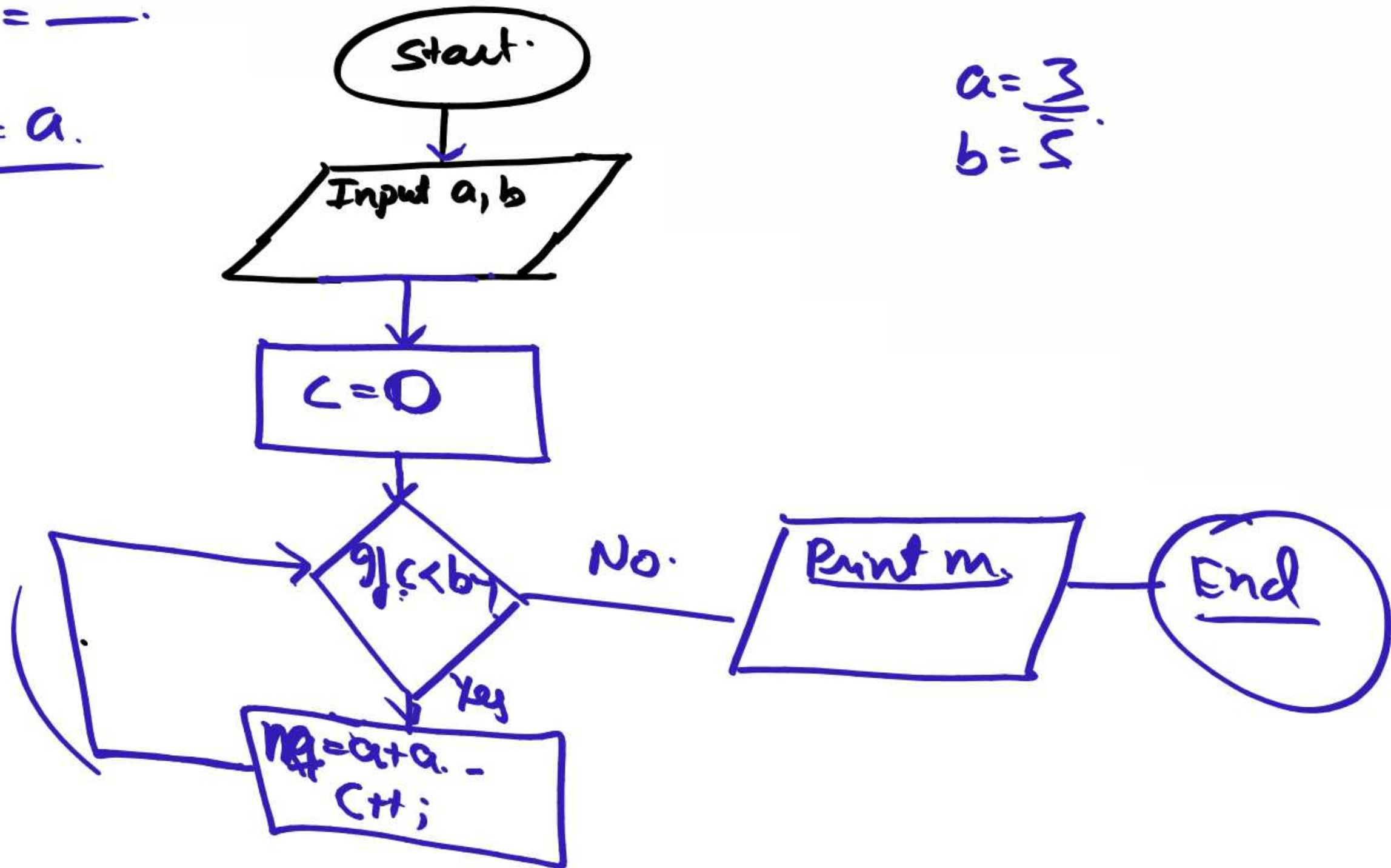
$$c = 2$$

$$m = 12$$

$$c = 3$$

$$m = 15$$

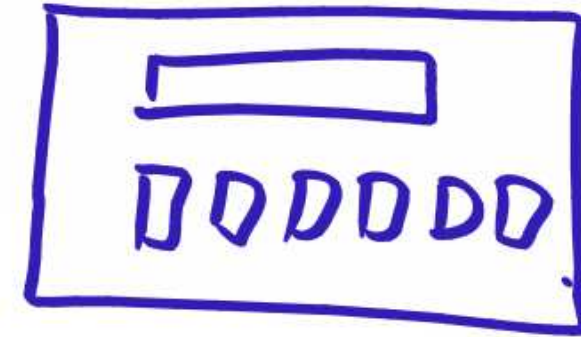
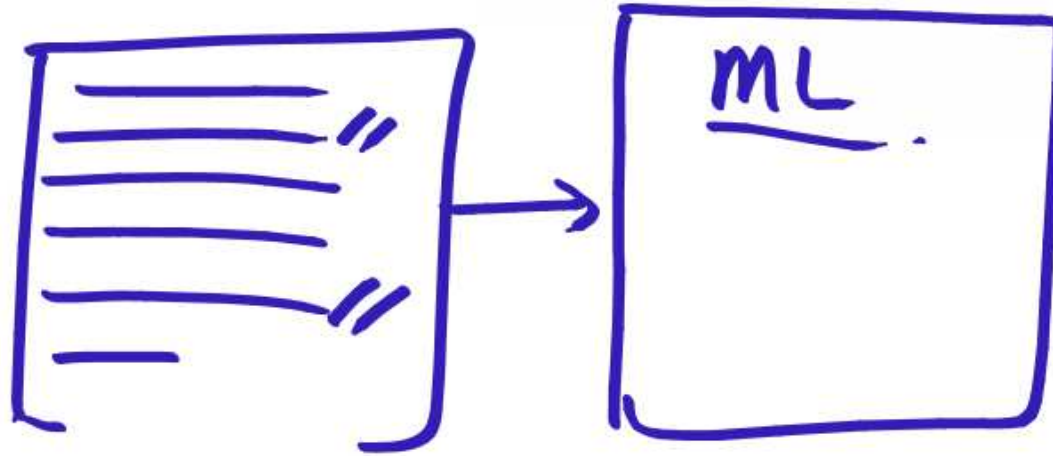
$$c = 4$$



Assembler.

Compiler, Interpreter,
Translator

C
C++
Java



8085

Assembly language

ADD a,b;

Object Oriented Language. → 3,00Ps

Object Data

Procedural Language.

PEAI
↓ ↓ ↓ ↓

↳ Algorithm based



2 mins Summary

Topic

Computational Thinking and Programming

THANK YOU